

Zad. 1

$$\sum_{n=1}^{\infty} \frac{1}{(2n-1)(2n+1)} = \sum_{n=1}^{\infty} \left(\frac{\frac{1}{2}}{2n-1} - \frac{\frac{1}{2}}{2n+1} \right) = \frac{1}{2} \sum_{n=1}^{\infty} \left(\frac{1}{2n-1} - \frac{1}{2n+1} \right) = \frac{1}{2} \lim_{n \rightarrow \infty} S_n = \frac{1}{2} \cdot 1 = \frac{1}{2}$$

$$\frac{1}{(2n-1)(2n+1)} = \frac{a}{2n+1} + \frac{b}{2n-1} = \frac{a(2n-1) + b(2n+1)}{(2n+1)(2n-1)}$$

$$1 = a(2n-1) + b(2n+1)$$

$$1 = n(2a+2b) + b - a$$

$$\begin{cases} 2a+2b=0 \\ b-a=1 \Rightarrow b=a+1 \end{cases}$$

$$2a+2(a+1)=0$$

$$4a = -2$$

$$a = -\frac{1}{2} \quad b = \frac{1}{2}$$

$$S_1 = a_1 = \frac{1}{1} - \frac{1}{3}$$

$$S_2 = a_1 + a_2 = \frac{1}{1} - \frac{1}{3} + \frac{1}{3} - \frac{1}{5}$$

$$S_3 = a_1 + a_2 + a_3 = \frac{1}{1} - \frac{1}{3} + \frac{1}{3} - \frac{1}{5} + \frac{1}{5} - \frac{1}{7}$$

⋮

$$S_n = a_1 + \dots + a_n = \left(1 - \frac{1}{2n+1} \right) \rightarrow 1$$

Zad. 2

$$\sum_{n=1}^{\infty} \frac{2^n}{7^n} = \sum_{n=1}^{\infty} \left(1 \cdot \left(\frac{2}{7} \right)^n \right) = \frac{2}{7} + \left(\frac{2}{7} \right)^2 + \left(\frac{2}{7} \right)^3 + \dots = \frac{\frac{2}{7}}{1 - \frac{2}{7}} = \frac{2}{5}$$

$$\sum_{n=2}^{\infty} \frac{1+2^n+3^n}{4^n} = \sum_{n=2}^{\infty} \left(\frac{1}{4^n} + \left(\frac{2}{4} \right)^n + \left(\frac{3}{4} \right)^n \right) = \sum_{n=2}^{\infty} \left(\frac{1}{4} \right)^n + \sum_{n=2}^{\infty} \left(\frac{2}{4} \right)^n + \sum_{n=2}^{\infty} \left(\frac{3}{4} \right)^n =$$

$$\begin{aligned} \sum_{n=1}^{\infty} \frac{2^n+5^n}{3^n} &= \sum_{n=1}^{\infty} \left(\left(\frac{2}{3} \right)^n + \left(\frac{5}{3} \right)^n \right) \rightarrow \infty \\ &= \frac{\frac{1}{16}}{1 - \frac{1}{4}} + \frac{\frac{1}{4}}{1 - \frac{1}{2}} + \frac{\frac{9}{16}}{1 - \frac{3}{4}} = \frac{1}{16} \cdot \frac{4}{3} + \frac{1}{4} \cdot \frac{2}{1} + \frac{9}{16} \cdot \frac{4}{1} = \\ &= \frac{1}{12} + \frac{1}{2} + \frac{3}{4} \end{aligned}$$

Zad. 3

Zbadaj zbieżność szeregu.

$$1) \sum_{n=1}^{\infty} \left(\frac{n^2}{2n^2+1} \right) \rightarrow \infty \quad \lim_{n \rightarrow \infty} a_n = \frac{1}{2} \neq 0$$

$$2) \sum_{n=1}^{\infty} \left(\frac{n+1}{3n+1} \right) \rightarrow \infty, \text{ bo } \lim_{n \rightarrow \infty} a_n = \frac{1}{3} \neq 0$$

$$3) \sum_{n=1}^{\infty} (\sqrt{n+2} - \sqrt{n})$$

$$a_n = \sqrt{n+2} - \sqrt{n} \cdot \frac{\sqrt{n+2} + \sqrt{n}}{\sqrt{n+2} + \sqrt{n}} = \frac{n+2-n}{\sqrt{n+2} + \sqrt{n}} = \frac{2}{\sqrt{n+2} + \sqrt{n}} \sim \frac{1}{\sqrt{n}} = b_n$$

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{2}{\sqrt{n+2} + \sqrt{n}} \cdot \sqrt{n} \rightarrow \frac{2\sqrt{n}}{\sqrt{n+2} + \sqrt{n}} \rightarrow$$

$$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} \rightarrow \infty$$

zKD
bo $\rho = \frac{1}{2} \leq 1$

$$\rightarrow \frac{2\sqrt{n}}{\sqrt{n}(\sqrt{1+\frac{2}{n}} + 1)} \rightarrow \frac{2}{2} \rightarrow 1 \in (0, \infty) \Rightarrow \sum_{n=1}^{\infty} a_n \rightarrow \infty$$

$$4) \sum_{n=2}^{\infty} \frac{\sqrt[3]{n}}{(n+1)\sqrt{n}}$$

$$a_n = \frac{\sqrt[3]{n}}{(n+1)\sqrt{n}} \sim \frac{n^{\frac{1}{3}}}{n \cdot n^{\frac{1}{2}}} = \frac{1}{n^{\frac{7}{6}}} = b_n \quad \sum b_n < \infty$$

zKD ($\rho = \frac{7}{6} > 1$)

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} \rightarrow \frac{n^{\frac{1}{3}}}{(n+1)n^{\frac{1}{2}}} \cdot \frac{n^{\frac{7}{6}}}{1} \rightarrow \frac{n^{\frac{3}{2}}}{(n+1)n^{\frac{1}{2}}} \rightarrow \frac{n}{n+1} \rightarrow 1$$

$$\sum \frac{n^{2n}}{7^n (n!)^2}, \sum \frac{n+3}{n^2+2n}, \sum \frac{3n^3+4n^2}{5n^5+7n^3}, \sum \frac{n \cdot \sqrt{n}}{3! n^5 (n^4+2n)}$$

$$\sum_{n=1}^{\infty} \left(\frac{n+1}{2n+1} \right)^n < \infty$$

K.C.

$$\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} \rightarrow \sqrt[n]{\left(\frac{n+1}{2n+1} \right)^n} \rightarrow \frac{n+1}{2n+1} \rightarrow \frac{1}{2} < 1$$

$$\sum_{n=1}^{\infty} \left(\frac{4n+1}{3n+7} \right)^n \rightarrow \infty$$

$$\sum \left(\frac{3^n \cdot n!}{n^n} \right)^{a_n} \rightarrow \infty$$

K. DA

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| \rightarrow \frac{3^{n+1} \cdot (n+1)!}{(n+1)^{n+1}} \cdot \frac{n^n}{3^n \cdot n!} = \frac{\cancel{3} \cdot 3 \cdot \cancel{(n+1)} \cdot \cancel{n!} \cdot n^n}{(n+1)^n \cdot \cancel{(n+1)} \cdot \cancel{3} \cdot \cancel{n!}} = 3 \cdot \left(\frac{n}{n+1} \right)^n =$$

$$\sum \frac{(\tan n)^{2n}}{3^n \cdot n^2}$$

$$= 3 \cdot \left(1 + \frac{-1}{n+1} \right)^n = \left. \begin{array}{l} x_n = \frac{-1}{n+1} \rightarrow 0 \\ y_n = n \rightarrow \infty \\ x_n \cdot y_n = \frac{-n}{n+1} \rightarrow -1 \end{array} \right\}$$

$$= 3 \cdot e^{-1} = \frac{3}{e} > 1$$

nie istnieje i

$$\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = \frac{\tan^2 n}{3 \cdot \sqrt[n]{n^2}} \rightarrow 1$$

$$\sum_{n=1}^{\infty} \sqrt{n} \cdot \sin \frac{1}{n^2+5}$$

$$a_n = \sqrt{n} \cdot \sin \left(\frac{1}{n^2+5} \right) \sim \sqrt{n} \cdot \frac{1}{n^2+5} = \frac{\sqrt{n}}{n^2+5} = b_n \text{ zbieżny, bo } \sum b_n \text{ jest podobny } \frac{1}{n^{\frac{3}{2}}}$$

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} \rightarrow \frac{\sqrt{n} \cdot \sin \frac{1}{n^2+5}}{1} \cdot \frac{n^2+5}{\sqrt{n}} \rightarrow \frac{\sin \frac{1}{n^2+5}}{\frac{1}{n^2+5}} \rightarrow 1$$

Zbadac' czy szereg jest

zbieżny bezwzględnie, warunkowo

czy rozbieżny

$$\sum_{n=1}^{\infty} \left((-1)^{n+1} \frac{1}{3n-1} \right)$$

Badam zbieżność bezwzględną,

czyli $\sum_{n=1}^{\infty} \frac{1}{3n-1} \sim \sum_{n=1}^{\infty} \frac{1}{n} \rightarrow \infty$

badany jest rozbieżny
bezwzględnie

Badam zbieżność warunkową

$$\sum (-1)^{n+1} \cdot \left(\frac{1}{3n-1} \right) = a_n$$

K. L.

$$\lim_{n \rightarrow \infty} \frac{1}{3n-1} = 0 \quad \forall n \in \mathbb{N} \quad \frac{1}{3n-1} > 0$$

$a_n \downarrow$ osiągnięte $\Rightarrow$ z KŁ szereg zbieżny
warunkowo

$$\sum_{n=1}^{\infty} (-1)^n \left(\frac{2n+1}{3n+1} \right)^n$$

Badam zbieżność bezwzględną,

czyli $\sum_{n=1}^{\infty} \left(\frac{2n+1}{3n+1} \right)^n < \infty$ $\nearrow a_n$

K. C

$$\lim_{n \rightarrow \infty} \sqrt[n]{a_n} \rightarrow \frac{2n+1}{3n+1} \rightarrow \frac{2}{3} < 1$$

Szereg zbieżny bezwzględnie

$$\textcircled{A} \quad a) \sum \frac{n^{2n}}{7^n (n!)^2}, \quad b) \sum \frac{n+3}{n^2+2n}, \quad c) \sum \frac{3n^3+4n^2}{5n^5+7n^3}, \quad d) \sum \frac{n \cdot \sqrt{n}}{\sqrt[3]{n^5} (n^4+2n)}$$

$$a) \sum \frac{n^{2n} // e_n}{7^n \cdot (n!)^2} \rightarrow \infty$$

K. DA

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| \rightarrow \frac{(n+1)^{2n+2}}{7^{n+1} [(n+1)!]^2} \cdot \frac{7^n \cdot (n!)^2}{n^{2n}} \rightarrow$$

$$\rightarrow \frac{(n+1)^{2n} \cdot \cancel{(n+1)^2} \cdot \cancel{7^n} \cdot \cancel{(n!)^2}}{\cancel{7^n} \cdot 7 \cdot \cancel{(n+1)^2} \cdot \cancel{(n!)^2} \cdot n^{2n}} \rightarrow$$

$$\rightarrow \frac{1}{7} \cdot \left(\frac{n+1}{2n} \right)^{2n} \rightarrow \begin{cases} x_n = \frac{n+1}{2n} \rightarrow \frac{1}{2} \\ y_n = 2n \rightarrow \infty \end{cases}$$

$$\rightarrow \frac{1}{7} \cdot e^\infty = \frac{e^\infty}{7}$$

$$x_n \cdot y_n = \frac{2n(n+1)}{2n} = n+1 \rightarrow \infty \rightarrow$$

$$b) \sum \frac{n+3}{n^2+2n} \rightarrow \infty$$

$$\frac{3}{2} - \frac{17}{3} = \frac{3}{6} - \frac{34}{6} = -\frac{25}{6} = -4\frac{1}{6}$$

$$c) \sum \frac{3n^3+4n^2}{5n^5+7n^3} = \left[\frac{1}{n^2} \right] \rightarrow 0$$

$$d) \sum \frac{n \cdot \sqrt{n}}{\sqrt[3]{n^5} (n^4+2n)} \rightarrow \frac{n^{\frac{3}{2}}}{n^{\frac{5}{3}} \cdot (n^4+2n)} \rightarrow \frac{n^{\frac{3}{2}}}{n^{\frac{17}{3}+2n^{\frac{4}{3}}}} \rightarrow 0$$